VEDAVYAS ROKALABOINA

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PERSONAL PROFILE

A motivated Aeronautical Engineering MSc graduate combining hands-on fabrication experience from SAE RC Aeromodelling competitions with advanced computational research skills. Successfully led project by designing and optimising a bio-inspired Blended-Wing-Body aircraft using ANSYS Fluent, validating complex CFD frameworks for peak aerodynamic efficiency. Proficient in SOLIDWOKS, ANSYS, CATIA, MATLAB, and XFLR, they excel in diagnosing engineering challenges and cross-functional collaboration. A detail-oriented professional adaptable to emerging technologies, delivering precision in design, simulation, and technical documentation within the aeronautical and aerospace sector.

WORK EXPERIENCE

TECHNICAL LEAD | ARDC IARE (Student Volunteer) **APR 2022 to FEB 2024**

Worked as a Technical Lead, overseeing aeronautical engineering projects in airframe design, aerodynamics, propulsion systems, and structural analysis. Led teams in designing, prototyping, and computational analysis of aerospace components using industry-standard software such as ANSYS, CATIA, SOLIDWORKS, XFLR, MATLAB, PYTHON, and AUTOCAD. Specialised in CFD simulations, FEM analysis, and flight performance evaluation, contributing to advanced research and development in the aeronautical sector.

ABHUDAY BHARAT PROJECTS PRIVATE LIMITED | TECHNICAL LEAD **MAR 2024 to SEP 2025**

Worked as a Technical Lead overseeing UAV and drone development, contributing to the full engineering lifecycle. This role involved designing critical components in SolidWorks and performing aerodynamic CFD and structural FEM analyses using ANSYS Fluent to optimise performance and reliability. Developed and tested avionics algorithms in MATLAB, ensuring seamless integration with flight-control systems. Working closely with multidisciplinary teams, helped to drive design decisions, troubleshoot technical challenges, and deliver high-quality prototypes aligned with project requirements.

EDUCATION

UNIVERSITY OF SOUTH WALES – Master`s degree **SEP 2024 – SEP 2025**

MSc Aeronautical Engineering.

Core modules covered airframe design, aerodynamics, propulsion, avionics, and computational analysis, integrated with engineering management and robotics. Key projects included designing a business jet simulated in Merlin and SolidWorks, alongside a significant dissertation on advanced engineering applications.

INSTITUTE OF AERONAUTICAL ENGINEERING - Bachelor`s degree **DEC 2020 – APR 2024**

Bachelor of Technology Aeronautical Engineering

Active participation in SAE Aero Modelling competitions and projects like Bi-Oblique Wing Design and Thrust Vectoring by Jet Vanes refined expertise in aircraft design and propulsion. Proficiency in ANSYS, CATIA, SOLIDWORKS and MATLAB enabled advanced simulations, strengthening technical acumen and problem-solving for engineering challenges.

PROJECTS AND ACCOMPLISHMENTS

SAE INTERNATIONAL AND SAE NATIONAL AERO DESIGN CHALLENGE **JAN 2022 – MAR 2023**

Served as Technical Lead in which CFD, FEA analysis and CAD were executed, resulting in securing 15th place at the SAE Aero International Competition (Florida, USA, 2023) and 3rd place at the SAE Aero National Competition (Chennai, India, 2022).

THRUST VECTOR CONTROL BY JET VANES **JUN 2023 – APR 2024**

Performed CFD analysis on six different exit nozzles (square, rectangle, triangle, rhombus, circle, hexagonal) designed in CATIA for a jet-vane thrust vectoring project. The rectangular nozzle achieved the greatest 1.8 Mach core length compared to the other shapes.

BUSINESS JET DESIGN, ANALYSIS AND SIMULATION **SEP 2024 – APR 2025**

Designed an 8-passenger business jet by calculating the Take-Off Weight (TOW). MATLAB determined dimensions and engine requirements based on TOW. A SolidWorks CAD model was created, analysed via SolidWorks Simulation, and flown in the Merlin simulator, demonstrating low errors during cruise, take-off, and landing.

INVESTIGATION OF THE AERODYNAMICS OF A BIO – INSPIRED BLENDED WING BODY

APR 2025 – SEP 2025

Designed and optimised a bio-inspired Blended-Wing-Body (BWB) aircraft using SolidWorks and ANSYS Fluent. Validated the CFD framework through rigorous mesh convergence and external benchmarking. Conducted a parametric study on winglet sweep angles, identifying best configuration that maximised the Lift-to-Drag ratio, confirming superior aerodynamic efficiency.

SKILLS

- CFD AND FEA ANALYSIS
- OPEN FOAM
- ANSYS FLUENT
- PYTHON
- MATLAB
- AUTOCAD
- PROFSCAN
- XFLR5
- CATIA V5
- SOLIDWORKS
- GASTURB
- PROJECT MANAGEMENT
- TIME MANAGEMENT
- PROBLEM – SOLVING & TROUBLESHOOTING
- ADAPTABILITY
- COMMUNICATION SKILLS

ACADEMIC REFERENCE

DR ROBERT WARREN – Lecturer in Aeronautical Engineering (USW Treforest campus)

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